Qing Li | Curriculum Vitae

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RESEARCH INTEREST

Reinforcement Learning and Post-Training for Generative AI, Diffusion-Based Policy Learning, Reliable Sequential Decision-Making and Reasoning under Uncertainty, Robust Temporal Representation, Multi-Agent Learning.

EXPERIENCE

Xinjiang University Urumqi, China

Student Researcher - Generative Models and Multi-Agent Reinforcement Learning *07/2025 - Present*

- **Diffusion-Based Generative Policy Learning:** Integrated the reverse-denoising process of a diffusion model into a Multi-Agent Proximal Policy Optimization (MAPPO) policy network, using generative modeling to produce adaptive actions from sparse and complex observations.
- **Multi-Agent Policy Optimization under Partial Observability:** Formulated cooperative sequential decision-making as a decentralized partially observable Markov decision process (Dec-POMDP) and developed a multi-agent reinforcement learning framework for coordinated policy learning under non-stationarity.
- **Implementation and Systematic Experimental Evaluation:** Implemented the learning framework in PyTorch and evaluated policy performance in obstacle-rich CFD environments against multiple MARL baselines, which investigating sparse rewards, information sharing, coordination, and robustness.

Student Researcher - Deep Reinforcement Learning *09/2024 - 06/2025*

- **GRPO-Style Reinforcement Learning and Policy Optimization:** Developed a task-specific Enhanced Group Relative Policy Optimization (EGRPO) method with decoupled clipping, dynamic sampling, and reward reshaping to improve training stability and sequential decision performance under partial observability.
- **Robust Temporal Representation:** Developed a lightweight Transformer encoder with a learnable State token to extract temporal dependencies from noisy and partially missing observation sequences, improving state representation for downstream policy decisions.
- **Structured and Event-Driven Decision-Making:** Designed a Prioritized Colored Petri Net (PCPN) with explicit states, guarded transitions, and priority rules to coordinate exploration, exploitation, obstacle avoidance, recirculation-zone escape, and terminal decisions through an interpretable event-driven execution process.

North University of China Tai Yuan, China

Undergraduate Research Assistant - Intelligent Device for Autonomous Rubik's Cube Solving *09/2018 - 06/2022*

- **Control System Implementation:** Designed, assembled, and debugged a pneumatic actuation-based control system for automated Rubik's Cube manipulation, integrating mechanical actuation and sequential control.

PROJECTS

- **Diffusion-Enhanced Multi-Agent Policy Learning:** Proposed a Diffusion Model-Enhanced Multi-Agent Proximal Policy Optimization (DM-MAPPO) framework for cooperative sequential decision-making under partial observability. Formulated the task as a Dec-POMDP and integrated diffusion-based reverse denoising into the MAPPO policy network for generative action-policy learning. Implemented the framework in PyTorch and evaluated it against MAPPO, MADQN, VDN, and QMIX in obstacle-rich dynamic CFD simulations, demonstrating improved average reward and search efficiency. Manuscript under review at Neurocomputing.
- **Event-Driven Reinforcement Learning with GRPO-Style Policy Optimization:** Developed an event-driven RL framework combining Transformer-based temporal feature extraction, EGRPO policy optimization, and PCPN for robust and interpretable decision-making under noisy and partially missing observations. Validated in CFD-simulated gas-dispersion environments, achieving a 14.4% increase in average reward and a 28.5% reduction in search steps over PPO. Manuscript accepted by Applied Soft Computing.

EDUCATION

Xinjiang University Urumqi, China

M.Res. in Control Science and Engineering; GPA: 3.59/4.00 *09/2024 – 06/2027 (Expected)*

- Selected Coursework: Machine Learning and Applications, Optimization Theory and Methods, Matrix Analysis.

North University of China Taiyuan, China

B.Eng. in Electrical Engineering and Intelligent Control; GPA: 3.32/4.00 *09/2018 – 06/2022*

- Selected Coursework: Advanced Mathematics, Linear Algebra, Complex Variables Functions & Integral Transforms, Probability Theory & Mathematical Statistics.

PUBLICATIONS

[1] **Qing Li**, Jie Yuan*, Li Zhu, Ningning Zhang, Erkenbieke Jia. "Autonomous odor source localization using event-driven deep reinforcement learning and transformer-based feature evaluation." *Applied Soft Computing*, 2026. DOI: /10.1016/j.asoc.2026.116781. [JCR Q1, IF: 7.8]

[2] **Qing Li**, Jie Yuan*, Ningning Zhang, Erkenbieke Jia. "Generative AI-enhanced multiagent deep reinforcement learning for cooperative odor source localization." *Neurocomputing*, 2026. Status: Under Review. [JCR Q1]

SKILLS

Technical Skills: Proficient in PyTorch, CFD ANSYS, C/C++, MATLAB, Origin, Visio, LaTeX, Office.

Languages: Chinese (native); English (IELTS 6.5).

HONORS & AWARDS

- Academic Scholarship 2024, 2025
- National Encouragement Scholarship (Top 3%) 2019, 2020

REFERENCES

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